

Midterm Project Instructions

Data Mining and Machine Learning

Objective:

Students are required to select a real-life dataset and apply one data mining or machine learning method covered in class. The goal is to understand the full data mining process, explain the chosen method, and present meaningful results.

Project Requirements

- Choose a real-life dataset from any domain, such as healthcare, business, education, social media, banking, shopping, transportation, or sports.
- Clearly explain the project's target or problem.
- Explain why you selected this dataset.
- Use a data mining or machine learning method taught in class.
- Explain why you selected this method and why it is suitable for the dataset/problem.
- Perform the main data mining steps and explain them clearly.
- Present the final output/results and explain the meaning of the results.

Suggested Methods

- Decision Tree
- k-Nearest Neighbors (k-NN)
- Weighted k-NN
- Radius-Based Neighbors
- Naïve Bayes
- Rule-Based Classifiers
- Random Forest

Project Structure

1. Project Title
2. Introduction and Project Goal
3. Dataset Description
4. Data Cleaning / Preprocessing
5. Selected Method
6. Why This Method Was Chosen
7. Python Code and Explanation
8. Results and Output
9. Conclusion

Required Data Mining Steps

- Load the dataset using Python (pandas).
- Check the dataset information and columns.
- Handle missing values if necessary.

- Prepare features (X) and target (y).
- Split data into training and testing sets.
- Train the selected machine learning model.
- Evaluate the model using suitable metrics.
- Explain the final output and prediction results.

Submission Instructions

- Submit a PPT presentation.
- Include Python code screenshots or Jupyter Notebook output.
- Presentation should be simple and clear.
- Students must explain their work during the presentation.

Evaluation Criteria

Criteria	Marks
Project idea and dataset	20
Correct use of method	25
Code and implementation	20
Explanation and analysis	20
Presentation and organization	15